BHAVISH RAI B

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Objective

Programming and building robots that can operate effectively in complex and real-world applications.

Education

Bachelor of Engineering in Robotics and Automation — 2021 - 2025
Sahyadri College of Engineering and Management, Mangaluru — *CGPA: 8.31/10.0*

Pre-University Education (PCMCs) — 2019 - 2021
Sri Rama Pre-University College, Kalladka — *Percentage: 86%*

Secondary Education (SSLC) — 2019
Govt. High School Beliyurukatte, Puttur — *Percentage: 84.64%*

Publications

CoMuRoS: A LLM-Driven Hierarchical Architecture for Adaptive Multi-Robot Collaboration. [Paper] [Website]
*Suraj Borate, **Bhavish Rai B**, Vipul Pardeshi and Madhu Vadali.* IROS 2025 Workshop - LeaPRiDE: Learning, Planning, and Reasoning in Dynamic Environments, Hangzhou, China, 2025.

CoMuRoS: LLM-Based Generalizable Hierarchical Task Planning and Execution for Heterogeneous Robot Teams with Event-Driven Replanning. [Paper] [Website]
*Suraj Borate, **Bhavish Rai B**, Vipul Pardeshi and Madhu Vadali.* ArXiv, 2025. Under Review at Frontiers in Robotics and AI.

Application of LLM Driven Robot Architecture for the Study of Robot Interventions in Making Task.
*Suraj Borate, **Bhavish Rai B***, Megha Bansal*, Harshil Shafi, Saloni Shinde, Aditi Kothiyal, and Madhu Vadali.*
Submitted to IEEE Transactions on Learning Technologies, 2026.

Integrated Task and Kinodynamic Path Planning for Heterogeneous Multi-Robot Formations
*Suraj Borate, **Bhavish Rai B**, and Madhu Vadali.*
Accepted for ICRA 2026 Workshop Scaling Compositional Intelligence for Multi-Agent Robotic Systems (SCI-MARS).
IEEE Robotics and Automation Letters (RA-L). *[Under preparation]*

Patents

A Multi-Robot Coordination System for Task Planning and Execution — *Patent Applied*
*Inventors: Suraj Borate, Madhu Vadali, and **Bhavish Rai B***
Filing Date: 07.05.2026

Professional Experience and Projects

Research Intern (Jan - May 2025), Program Assistant - I (June 2025 - Present)
Indian Institute of Technology, Gandhinagar

- Contributed to design and code for ROS2 architecture of Collaborative Multirobot System(CoMuRoS) [Website]
- Contributed to finetuning of prompts for VLM and LLM for event detection and Task Manager of CoMuRoS. [Video]
- Contributed develop a Navigation Manager that converts high-level natural language tasks into metric goal poses and implemented path planners for both single-robot and multi-robot navigation in heterogeneous robot systems. [Video]

- Vision+PID based Controllers for mobile robots, implemented control for holonomic robot and kanayama controller for differential drive robot. [Video]
- Curated Dataset of Scenario Configurations and Tasks for testing generalization of CoMuRoS across Scenarios, Robots, Tasks and LLMs.
- Implemented ChoiRbot library for formation control of mobile robots on Turtlebot3 Hardware platform. [Video]
- Implemented position control on actual Unitree Go2 robot. [Video]
- Developed multiple Gazebo environments like Hospital, Disaster Relief and Restaurant for testing of CoMuRoS. [Video]
- Fabricated 3 Lerobot Arms. Performed data collection, finetuning of Nvidia Groot VLA and successful testing of pick and place of objects using VLA. [Video]
- Built bi-manual social tutorbot from scratch using Lerobot and LLMs, this robot could communicate through voice commands and guided students in assembly tasks using the CoMuRoS architecture as backbone. [Video]
- Contributed work to a talk in ROSCON 2025, multiple research publications and one patent application. [Video]
- Implemented pick and place on X-arm Manipulator using MoveIt. [Video]
- Implemented Nav2 on Turtlebot3.
- Implemented a Navigation Manager module that combines Task Planning of CoMuRoS with Path Planning.

Research Intern - Robotic Puppy Dog Project (October 2023 - November 2023)
Technical Career Education, Adyar, Mangaluru
- Designed and developed a quadruped robotic platform with actuated joints and integrated sensors
- Prototyped the physical leg fixture of the robot dog with servo motors

Technical Projects

CoMuRoS: Collaborative Multi-Robot System IIT Gandhinagar
- Developed a generalizable hierarchical architecture for LLM-driven task planning and event-driven replanning in heterogeneous multi-robot teams, unifying centralized deliberation with decentralized execution
- Implemented Task Manager LLM for natural-language goal interpretation and dynamic task allocation using static rules and contextual information (task history, robot status, events) [Video]
- Designed local robot LLMs that compose executable Python code from primitive ROS2 skills, with onboard VLM-based perception for continuous event monitoring and classification [Video]
- Achieved 90% collaborative object recovery success (9/10), 100% coordinated transport (8/8), and 100% human-assisted recovery (5/5) in hardware experiments
- Demonstrated autonomous recovery from disruptive events, filtering of irrelevant distractions, and emergent human-robot cooperation through runtime replanning [Video]
- Developed using ROS2 Humble with Python, simulated in Gazebo Classic and Ignition Fortress [Video]

Integrated Task and Kinodynamic Path Planning for Heterogeneous Multi-Robot FormationsIIT Gandhinagar
- Developed a Navigation Manager that converts high-level natural language tasks into metric goal poses understandable by path planners.
- Implemented path planners for both single-robot and multi-robot navigation in heterogeneous robot systems. [Video]

An Agentic AI Architecture for Robot-Assisted Hands-on Learning. IIT Gandhinagar
- Developed an LLM-driven Bimanual TutorBot capable of interacting with users through speech and facial expressions [Video]
- Integrated two LeRobot arms as bimanual manipulators for interactive tutoring tasks
- Used a display screen to show robot's facial expressions for improved human-robot interaction
- Implemented CoMuRoS architecture by treating the two robotic arms as a multi-robot system
- Enabled real-time assistance for students performing assembly tasks
- Designed voice-based interaction using microphone input for student queries
- Implemented Socratic-style guidance through verbal responses and action-based demonstrations
- Enabled object picking and visual hint demonstration to assist task completion

3-DOF Robotic Arm Manipulator

- Designed a 3-degree-of-freedom robotic arm using SolidWorks
- Implemented forward and inverse kinematics for precision control and trajectory planning
- Developed control algorithms for pick-and-place automation tasks

Poultry Patrol Robot

- A robot that detects and marks diseased chickens in poultry farms using computer vision and ML [Video]
- Designed a differential drive mobile robot in SolidWorks (2D/3D) for autonomous poultry farm surveillance, fabricated using aluminum sheet metal, 3D-printed PLA parts, and black acrylic panels
- Integrated Raspberry Pi 5 with YOLOv5 for real-time diseased chicken detection via webcam, and ESP32 for distributed sensor data acquisition and wireless telemetry
- Implemented motor control via L298N drivers and a servo-actuated color sprayer mechanism to autonomously mark identified diseased birds
- Built on ROS2 with dedicated nodes for image capture, ML inference, and motor control, enabling autonomous navigation and obstacle avoidance in farm environments
- [Report]

Automated Home Security System

- Developed IoT-based surveillance system with computer vision for intruder detection
- Integrated Telegram bot API for real-time alerts and remote monitoring capabilities
- Implemented motion detection and image capture using Raspberry Pi and camera module
- [Report]

Automated Farm Water Pump System

- Designed an IoT-based smart irrigation system for agricultural use, automating water pump control based on real-time soil moisture data
- Integrated soil moisture sensor and motor driver with ESP32, streaming live sensor data to Adafruit IO dashboard for remote monitoring
- Implemented event-driven email notifications via IFTTT, alerting farmers instantly on every pump state change without manual checking

Design and Analysis of Helical Spring

- Designed a helical spring (d=14mm, D=70mm, 10 active coils) in SolidWorks and performed nonlinear FEA simulation for shear stress and deflection under a 3768.7N load
- Validated FEA results against hand-calculated theoretical values using Wahl's correction factor, achieving 1.14% error in deflection (35mm vs 35.4mm)
- Developed a Python script to parametrically analyze the effect of mean diameter on induced shear stress, plotting results across a range of 55–85mm
- [Report]

Technical Skills

Programming Languages: Python, C++, C, I can adapt to new language quickly.
Robotics & Simulation: ROS2 Humble, Gazebo Classic, Ignition Fortress, RVIZ
CAD & Design: SolidWorks, Fusion 360
Embedded Systems: Arduino, ESP32, ESP8266, Raspberry Pi, Jetson Nano Orin NX
Development Tools: Git, GitHub, VS Code, Linux (Ubuntu 22.04)
Prototyping: 3D Printing, Laser Cutting, Soldering
Soft Skills: Problem Solving, Team Collaboration, Project Management, Rapid Prototyping, Event Organization

Robot Platforms Experience

Custom Platforms: Poultry Robot
Quadruped Robots: Unitree GO2 (Edu)
Wheeled Mobile Robots: TurtleBot4, TurtleBot3 Burger, TurtleBot3 Waffle Pi, LeKiwi
Manipulators: SO101 Arm,TurtleBot3 Waffle Pi with Manipulator, Open Manipulator X

Achievements & Leadership

SAE AEROTHON 2024: Secured All India Rank 22 in national-level aerospace competition
Mattarme Robo Race: First Place in autonomous robot racing competition
Aerophilia 2023: Organized techno-cultural fest as core team member (December 2023)
Team Challengers Club: Contributed to team robotics projects while mentoring junior members across a 3-year period (February 2022 - January 2025)

Certifications

ROS2 for Beginners Level 1 & 3 - ROS2 fundamentals to advanced robotics (April 2024, August 2024) [Certificate 1] [Certificate 2]
OpenCV Bootcamp - Computer vision and image processing fundamentals (May 2024) [Certificate]